

Determine Surface Area of Pyramids

Lesson 5-8

Name: Type your name

Date: Today's date

Class: Class period

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can find the surface area of a pyramid by adding the base area and the lateral faces.

Language Objective: I can explain my work using the words pyramid, slant height, lateral face, and base.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Surface Area of Pyramids

Pyramid

Slant height

Lateral face

Base

Apex

Lateral area



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The total area of a pyramid's faces, found with $SA = B + \text{lateral area}$, is the

2

A solid with a polygon base and triangular faces meeting at one point is a

3

The height of a triangular face of a pyramid, measured along the slanted side, is the

4

A triangular side face of a pyramid is a

5

The polygon on the bottom of a pyramid is the

- 6 The single top point where the triangular faces of a pyramid meet is the

.

- 7 The combined area of all the side faces, not counting the base, is the

.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How much will the glass for all 4 triangular sides cost? (The base is open — no glass needed.)?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Surface Area of Pyramids** — The total area of a pyramid's base and triangular faces. Formula: $SA = B +$ lateral area.
Área de superficie de pirámides — El área total de la base y las caras triangulares de una pirámide. Fórmula: $SA = B +$ área lateral.

- ☐ **Pyramid** — A solid with a flat bottom and triangle sides that meet at one point on top.
Pirámide — Un sólido con un fondo plano y lados triangulares que se unen en un punto arriba.

- ☐ **Slant height** — The height of a side triangle, measured along its slanted face.
Altura inclinada (apotema) — La altura de un triángulo lateral, medida por su cara inclinada.

- ☐ **Lateral face** — A triangle side of a pyramid, not the bottom.
Cara lateral — Un lado triangular de una pirámide, no el fondo.

- ☐ **Base** — The flat bottom of a pyramid.
Base — El fondo plano de una pirámide.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

Each triangular face has area $= \frac{1}{2} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ ft}^2$. **Four faces:** $4 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ ft}^2$. **Cost:** $\underline{\hspace{1cm}} \times \$12 = \$\underline{\hspace{1cm}}$.

Di tu respuesta primero: Mi respuesta es $\underline{\hspace{1cm}}$ porque $\underline{\hspace{1cm}}$.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Surface area of a pyramid is the base area plus the areas of all the triangular lateral faces.

Evidence: Students explain a square pyramid's surface area = base area + the four triangular lateral faces, and that you add every face together.

Reasoning: This evidence connects the definition of surface area of pyramids to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Each triangular face has area** = $\frac{1}{2} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ **ft². Four faces:** $4 \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ **ft². Cost:** $\underline{\hspace{1cm}} \times \$12 = \$\underline{\hspace{1cm}}$.

- **My answer is** $\underline{\hspace{1cm}}$.

Mi respuesta es $\underline{\hspace{1cm}}$.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is** $\underline{\hspace{1cm}}$.

Mi afirmación es $\underline{\hspace{1cm}}$.

- **I know because** $\underline{\hspace{1cm}}$.

Lo sé porque $\underline{\hspace{1cm}}$.

- **So** $\underline{\hspace{1cm}}$.

Entonces $\underline{\hspace{1cm}}$.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.